Root-local Gram secants and the nonduplicating feedback boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

R-093 reduced the remaining regular one-shot problem to a coefficient- unconditioned root-local estimate for the complete R-079 future packet. It suggested the scale 2^{j-4k} for a probability root j and a later physical control shell $k > j$. This note determines exactly where that scale is valid, proves the resulting centered Gram-secant estimate without assuming selector independence, and audits whether it closes the complete packet.

There are four positive conclusions. First, the literal positive quadratic Gram-curvature atom does have the scale 2^{j-4k}. Second, the full centered Gram secant contains a mixed term of the weaker scale 2^{j-2k}, but its weighted Hardy sum is still form-subcritical: its mixed budget is $X^{1/2}Y^{1/6}$ and has Young slack 1/3. The proof uses product-space L^2–L^6 interpolation, not a pathwise Bernstein estimate that would ask for an unavailable higher selector moment. Third, the isolated fresh derivative channel has budget $X^{1/2}Y^{1/3}$ and slack 1/6. Fourth, the combined value-innovation plus heat coefficient acting on the accumulated control derivative costs only $O(2^{-3j_0}X)$ when completed against a fixed fraction of the already retained derivative-feedback square.

The hostile nonduplication audit is decisive. These estimates do not yet embed themselves into the complete R-079 packet. After the legitimate partial-square payment, the coefficient/derivative cross, the Gaussian value innovation and heat residual, the R-086 $T_G^{>}$ coefficient-dominant branch, and the R-093 conditional mean debts remain. The square, covariance trace, backward heat, R-063 forest, low term, and paid subtraction must each occur once in the missing embedding identity. Thus R-094 is a genuine analytic advance and a sharper boundary, not a proof of H_N, REG, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, a measure, or Sector A. Tier stays T4.

The result is for the regular mutually orthogonal strict-past one-shot class. A two-step cancelling revisit shows why its per-shell sixth-moment step cannot be summed absolutely for arbitrary temporal revisits. The complete signed endpoint remains the required object there.

2 Regular hypotheses and budgets

Work on the normalized product space $S = \Omega \times \mathbb{T}^3$ and write $\|\cdot\|_p$ for $L^p(S)$. The probability filtration is $(\mathcal{F}_j)_{j\geq j_0-1}$. The fresh Gaussian value root is g_j, its spatial derivative is $d_j = Dg_j$, and

$$\mathcal{Q}_j = d_j \otimes d_j - \Delta\Gamma_j. \tag{2.1}$$

The accepted fourth-order covariance scaling gives constants independent of the terminal cutoff such that

$$\|d_j\|_6 \leq \kappa_1 2^{j/2}, \qquad \|\mathcal{Q}_j\|_3 \leq \kappa_Q 2^j, \qquad \operatorname{tr}\Delta\Gamma_j \leq \kappa_\Gamma 2^j, \tag{2.2}$$

and the value covariance satisfies

$$\operatorname{tr}\Sigma_j^{\mathrm{val}} \leq \kappa_0 2^{-j}. \tag{2.3}$$

Let M be the complete production frame. Its linear rows are linear and its normalized rational row has bounded first derivative by R-092. Consequently, after the common target-value heat if present,

$$\|M(z+a)-M(z)\| \le L_M|a|, \qquad |M(z)| \le L_M(1+|z|). \tag{2.4}$$

For $B(z)=M(z)Q_{\mathrm{II}}M(z)^T$, Gram expansion gives

$$|B(z+a)-B(z)| \le L_B\{(1+|z|)|a|+|a|^2\}. \tag{2.5}$$

The regular controls are

$$a_k = K_k h_k, \qquad a_k \in \mathcal{F}_{k-1}, \tag{2.6}$$

with mutually orthogonal physical shells and

$$\|a_k\|_2 \le \kappa_K 2^{-2k} q_k^{1/2}, \qquad \|Da_k\|_2 \le \kappa_D 2^{-k} q_k^{1/2}, \qquad q_k = \mathbb{E}\|h_k\|^2_{\mathcal{H}_k}. \tag{2.7}$$

Put

$$X=\sum_k q_k, \qquad v_k = \|a_k\|_6, \qquad A_{>j}=\sum_{k>j} a_k. \tag{2.8}$$

The regular Littlewood–Paley square function and the accepted base sixth moment give

$$\sum_k v_k^6 \le C_{\mathrm{LP}}\{1+Y\}, \qquad Y=\mathbb{E}\|Z_*\|_6^6. \tag{2.9}$$

This is the only sixth-moment input below. No L^p source moment with $p>2$ is assumed.

3 The factor audit: quadratic and mixed scales differ

Theorem 3.1 (quadratic Gram-curvature atom)

Let E_k be a frame increment with $\|E_k\| \le L_M|a_k|$ and define

$$R^{(2)}_{j,k} = \frac{1}{2}\mathbb{E}\int\left\{|Q_{\mathrm{II}}^{1/2}E_k^T d_j|^2 - (E_kQ_{\mathrm{II}}E_k^T):\Delta\Gamma_j\right\}, \qquad j<k. \tag{3.1}$$

Then

$$\boxed{R^{(2)}_{j,k} \ge -C_{\mathrm{curv}}2^{j-4k}q_k.} \tag{3.2}$$

Indeed, retain the first square and bound only the trace by (2.2), (2.4), and (2.7). No independence between E_k and root j is used. Summing gives

$$\sum_{j\ge j_0}\sum_{k>j} R^{(2)}_{j,k} \ge -C2^{-3j_0}X. \tag{3.3}$$

The dimensionless triangular kernel has the exact diagnostic

$$\sum_{j\ge j_0}\sum_{k>j} 2^{j-4k} = \frac{8}{105}2^{-3j_0}. \tag{3.4}$$

The conclusion cannot be promoted to the mixed Gram term. For $g=\sqrt{\gamma}\,\xi$, $H_2=\xi^2-1$, $m_0=1$, and $m_1=1-\varepsilon t H_2$, Gaussian moments give

$$\frac{1}{2}\mathbb{E}[(m_1^2-m_0^2)(g^2-\gamma)] = -2\gamma\varepsilon t + 4\gamma\varepsilon^2 t^2. \tag{3.5}$$

After adding ρt^2, its minimum is

$$-\frac{\gamma^2\varepsilon^2}{\rho+4\gamma\varepsilon^2}<0. \tag{3.6}$$

The linear secant therefore scales as 2^{j-2k}; only its post-Young square has scale 2^{2j-4k}. The latter remains summable, with

$$\sum_{j\ge j_0}\sum_{k>j}2^{2j-4k}=\frac{4}{45}2^{-2j_0}. \tag{3.7}$$

Equations (3.2) and (3.5) are the factor correction to the R-093 prototype.

4 The complete centered Gram secant is form-subcritical

Theorem 4.1 (regular centered Gram-secant bound)

Let $Z_j = Z_* - A_{>j}$ and define

$$\mathcal{S}_{\rm sec}=\sum_{j=j_0}^{J}\left|\mathbb{E}\int[B(Z_*)-B(Z_j)]:\mathcal{Q}_j\right|. \tag{4.1}$$

Then, uniformly in J,

$$\boxed{\mathcal{S}_{\rm sec}\le C_{\rm sec}2^{-j_0}X^{1/2}(1+Y)^{1/6}.} \tag{4.2}$$

Consequently, for all $\eta,\zeta\in(0,1]$,

$$\boxed{\mathcal{S}_{\rm sec}\le \eta X+\zeta Y+C_{\eta,\zeta},} \tag{4.3}$$

where one may take

$$C_{\eta,\zeta}\le C\{1+\eta^{-1}+\eta^{-3/2}\zeta^{-1/2}\}. \tag{4.4}$$

Proof. By (2.2), (2.5), and Holder on the product space,

$$\mathcal{S}_{\rm sec}\le C\sum_j 2^j\left\{(1+\|Z_*\|_6)\|A_{>j}\|_2+\|A_{>j}\|_3^2\right\}. \tag{4.5}$$

For the linear part, Tonelli and (2.7) give

$$\sum_j 2^j\|A_{>j}\|_2\le\sum_k\|a_k\|_2\sum_{j<k}2^j\le C\sum_k 2^{-k}q_k^{1/2}\le C2^{-j_0}X^{1/2}. \tag{4.6}$$

For the quadratic and cross-shell part, let $b_k=\|a_k\|_3$. Under the change $c_k=2^{k/2}b_k$, the tail map is convolution by $2^{-n/2}\mathbf{1}_{n\ge1}$. Its ℓ^1 norm is $1+\sqrt{2}$. Therefore

$$\sum_j 2^j\left(\sum_{k>j}b_k\right)^2\le(3+2\sqrt{2})\sum_k 2^k b_k^2. \tag{4.7}$$

The load-bearing interpolation is on S, not pathwise:

$$b_k^2=\|a_k\|_3^2\le\|a_k\|_2\|a_k\|_6. \tag{4.8}$$

Using (2.7), Holder in the shell index, and (2.9),

$$\begin{aligned}\sum_k 2^k b_k^2 &\le \kappa_K \sum_k 2^{-k} q_k^{1/2} v_k \\ &\le \kappa_K X^{1/2} \left(\sum_k 2^{-2k} v_k^2\right)^{1/2} \\ &\le C 2^{-j_0} X^{1/2} \left(\sum_k v_k^6\right)^{1/6} \\ &\le C 2^{-j_0} X^{1/2} (1+Y)^{1/6}.\end{aligned} \tag{4.9}$$

This proves (4.2). The mixed power $X^{1/2}Y^{1/6}$ has slack $1-1/2-1/6=1/3$, and multifactor Young proves (4.3)–(4.4). ☐

The theorem is expectation-inside and tolerates arbitrary strict-past selectors in the declared regular class. A spatial Bernstein estimate for each selected shell would incorrectly demand a higher probability moment.

5 Two genuine one-use payments

Lemma 5.1 (isolated fresh derivative form)

For the isolated form

$$\mathcal{T}_{\text{fresh}} = \sum_{j<k} \left| \mathbb{E} \int (Dg_j)^T B(Z_*) Da_k \right|, \tag{5.1}$$

the quadratic growth of the production Gram gives $\|B(Z_*)\|_3 \le C(1+Y^{1/3})$. Holder with exponents $(6,3,2)$, (2.2), and (2.7) yield

$$\boxed{\mathcal{T}_{\text{fresh}} \le C 2^{-j_0/2} X^{1/2} (1+Y^{1/3}).} \tag{5.2}$$

Here $\sum_{j<k} 2^{j/2} \le (1+\sqrt{2}) 2^{k/2}$. Thus

$$\mathcal{T}_{\text{fresh}} \le \eta X + \zeta Y + C\{1+\eta^{-1}+\eta^{-3}\zeta^{-2}\}. \tag{5.3}$$

The mixed power has slack 1/6. This is a valid analytic form bound; it is not yet an identification with all of the adapted R-079 derivative-feedback term, whose coefficient is evaluated at $U+A^*$.

Theorem 5.2 (combined value–heat control prefix)

Let $\overline{M}_j = P_{\Sigma_{j+1:J}} M$ and let W_{j-1} be the predictable translated background. Define

$$\delta_j M = \overline{M}_j (W_{j-1}+g_j) - \overline{M}_{j-1}(W_{j-1}). \tag{5.4}$$

This is exactly the value innovation plus heat compensator in R-079 (6.8). It is conditionally centered. Gaussian Poincare, the common-coupling Lipschitz bound, and (2.3) give

$$\mathbb{E}_{j-1} |\delta_j M|^2 \le L_M^2 \kappa_0 2^{-j}. \tag{5.5}$$

Let

$$H_j = D(A^{(j)} - A^0) = \sum_{k\le j} Da_k, \qquad x_j = Q_{\text{II}}^{1/2} (\delta_j M)^T H_j. \tag{5.6}$$

Then

$$\sum_j \mathbb{E}\|x_j\|^2 \le C \sum_j 2^{-j} \mathbb{E}\|H_j\|_2^2. \tag{5.7}$$

Without physical orthogonality, the prefix matrix $T_{jk} = 2^{-j/2}2^{-k}\mathbf{1}_{k\le j}$ has

$$\|T\|_{\mathrm{HS}}^2 = \sum_{j\ge j_0}\sum_{k=j_0}^{j} 2^{-j-2k} = \frac{16}{7}2^{-3j_0}. \tag{5.8}$$

Hence

$$\boxed{\sum_j \mathbb{E}\|x_j\|^2 \le C\frac{16}{7}2^{-3j_0}X.} \tag{5.9}$$

Mutually orthogonal shells improve the dimensionless constant to 2.

For any derivative-feedback vector y_j and $0<\theta<1$,

$$\mathbb{E}\langle x_j, y_j\rangle + \frac{\theta}{2}\mathbb{E}\|y_j\|^2 \ge -\frac{1}{2\theta}\mathbb{E}\|x_j\|^2. \tag{5.10}$$

Thus this prefix uses only a declared fraction θ of the retained feedback square and costs $C\theta^{-1}2^{-3j_0}X$. The fraction $1-\theta$ remains visible for the successor packet.

6 Exact nonduplication and the surviving packet

Theorem 6.1 (exact square-allocation split)

In R-079 (5.2), write the future innovation as

$$i_j = u_j + y_j, \tag{6.1}$$

where u_j is the R-079 future coefficient line and y_j is the future derivative-control line. Let $\delta_j^{\mathrm{D}} = P_j - P_{j-1}$ denote the Doob difference, distinct from the spatial Gaussian derivative $d_j = Dg_j$, and split

$$\delta_j^{\mathrm{D}}\mathcal{J}_j = p_j + x_j, \tag{6.2}$$

where x_j is (5.6) and p_j contains the remaining fresh-Gaussian, Gaussian-background, and heat/current pieces. Put

$$\tau_j = \frac{1}{2}\mathbb{E}\int(\mathcal{B}_* - \mathcal{B}_j) : \Delta\Gamma_j. \tag{6.3}$$

For every $0<\theta<1$, the future block has the exact identity

$$\boxed{\begin{aligned}\mathfrak{I}_j = {}& \left[\mathbb{E}\langle p_j + x_j, u_j\rangle + \frac{1}{2}\mathbb{E}\|u_j\|^2 - \tau_j\right]\\ & + \mathbb{E}\langle p_j + u_j, y_j\rangle + \frac{1-\theta}{2}\mathbb{E}\|y_j\|^2\\ & + \left[\mathbb{E}\langle x_j, y_j\rangle + \frac{\theta}{2}\mathbb{E}\|y_j\|^2\right].\end{aligned}} \tag{6.4}$$

It is merely the expansion of R-079 (3.4), but it is the necessary nonduplication ledger. The last line is paid by Theorem 5.2. The first two lines remain one coupled object. In particular, $\langle u_j, y_j\rangle$, the coefficient square, the trace, and the remaining feedback square occur exactly once.

This exposes the error in a tempting assembly: Theorem 4.1 cannot consume the whole coefficient square and then permit a second completion of the full $u_j + y_j$ square. If $y_j = -u_j$, then

$$\langle u_j, y_j\rangle + \frac{1}{2}\|y_j\|^2 = -\frac{1}{2}\|u_j\|^2, \tag{6.5}$$

so the leftover has no standalone sign.

Under the R-086 spatial decomposition, a conservative sufficient successor target, conditional on the missing once-only rootwise embedding, is the reduced-square coefficient-dominant packet

$$\boxed{\mathcal{P}^{>}_{R,\theta} = T^{>}_{Q} + T^{>}_{G} + \frac{1-\theta}{2} c^{T}\overline{B}(U+a)c,} \tag{6.6}$$

where $c = D(A^{*} - A^{0})$ is the R-086 spatial derivative variable and is distinct from the R-079 coefficient line u_j in (6.1). The target retains the complete Wick trace and R-063 lower-chaos forest. A bound for the full centered secant does not by itself bound $T^{>}_{Q}$ after pieces known only from below have been subtracted. More importantly,

$$T^{>}_{G} = \int_0^1 (1-t) \sum_{r} \sum_{\substack{|m-n|\le L_0 \\ m>r+L_0}} \int \langle \Delta_n G, (\Delta_m K_t)[H_r] \rangle \, dx \, dt \tag{6.7}$$

has a high, adapted, same-root coefficient. Boundedness is not a multiplier theorem for (6.7). The predictable base-current heat identity also does not apply to the terminal current $\mathcal{J}_*$ when future controls depend on root j.

Finally, the R-093 one-reveal normal form leaves, in general,

$$-q_B^{T}\overline{A}^{-1} q_B - |\mathbb{E}_{\mathcal{F}} \Theta_R(B)^{1/2} z|^2. \tag{6.8}$$

The centered even-reveal specialization makes these vanish, but a regular multistep root has a past background and an adapted coefficient. These mean debts must either be estimated or cancelled in the same missing embedding.

The exact successor is therefore the uniform lower bound

$$\mathbb{E}\left[\mathcal{P}^{>}_{R,\theta} + \text{complete Wick trace and lower-chaos forest}\right] \ge -\eta \mathbb{E}X - \zeta \mathbb{E}Y - C_{\eta,\zeta,e}, \tag{6.9}$$

for one fixed $0 < \theta < 1$, together with the rootwise identity connecting (6.4) to (6.6), the backward heat, low, and paid terms. Equivalently, it is the reduced-square version of R-092 (10.10) for the full derivative-feedback and Jensen packet.

7 Audit of the R-093 static exponent table

The two zero-slack rows displayed in the optional R-093 BG table were historical coarse bounds, not surviving exact atoms. The unshifted A^2DA current is paid by R-074/R-075 with

$$RX^{1/2}Y^{1/3}, \qquad s = \frac{1}{6}, \qquad p_R = 6, \tag{7.1}$$

and the unshifted A^3DA graph term is sharpened by R-076 to

$$RX^{2/5}Y^{8/15}, \qquad s = \frac{1}{15}, \qquad p_R = 15. \tag{7.2}$$

Thus neither old zero-slack row is the present obstruction. This does not prove the R-093 static criterion: its hypothesis is a complete expectation- inside shifted reconstruction. Equations (6.6)–(6.9), rather than model moments, are the current analytic gate.

8 Revisit and absolute-summation boundary

Let $E \in \mathcal{F}$ have probability p. Choose a unit source mode $e_{\mathcal{H}}$ for a nonzero smoothing range K and put $\phi = Ke_{\mathcal{H}} \neq 0$. Two later uses of that same range may take

$$h_1 = p^{-1/2}\mathbf{1}_E e_{\mathcal{H}}, \quad h_2 = -h_1, \qquad a_1 = Kh_1 = p^{-1/2}\mathbf{1}_E \phi, \quad a_2 = -a_1. \tag{8.1}$$

Smooth bounded approximations preserve the scaling. Their source cost is

$$q_1 + q_2 = \mathbb{E}\{\|h_1\|_{\mathcal{H}}^2 + \|h_2\|_{\mathcal{H}}^2\} = 2, \tag{8.2}$$

and the terminal displacement is zero, but

$$\|a_1\|_6^6 + \|a_2\|_6^6 = 2\|\phi\|_{L^6(\mathbb{T}^3)}^6 p^{-2} \longrightarrow \infty. \tag{8.3}$$

Therefore the regular estimate (2.9) cannot be applied separately to every revisit increment. This is a no-go for absolute per-revisit secant summation, not a counterexample to the complete signed action, in which the two increments cancel.

9 Proof and failure evidence map

$$\begin{array}{c}
\text{R-093 root-local prototype} \longrightarrow \boxed{\text{Theorem 3.1: } 2^{j-4k} \text{ only for quadratic curvature}} \\
\downarrow \\
\boxed{\text{Theorem 4.1: full centered secant is form-subcritical}} \\
\downarrow \\
\boxed{\text{Theorem 5.2: value–heat control prefix is one-use}} \\
\downarrow \\
\boxed{\text{Theorem 6.1: exact fractional square allocation}} \\
\downarrow \\
\text{open: rootwise embedding} + T_G^{>} + \text{mean debts} \\
\downarrow \\
H_N \ \longrightarrow \ \text{OVERLAP}_{\text{src}} \equiv q = 10/9 \text{ Nelson.}
\end{array} \tag{9.1}$$

The failed branches are retained rather than erased:

- promoting 2^{j-4k} to the complete mixed secant fails by (3.5);
- reusing the positive feedback square fails by (6.5);
- treating the unshifted $X^{1/2}Y^{1/3}$ form as the adapted $T_G^{>}$ multiplier estimate fails because (6.7) has a shifted same-root high coefficient;
- termwise revisit sixth-moment summation fails by (8.1)–(8.3);
- the two R-093 zero-slack rows are superseded coarse estimates, but their repair does not manufacture the missing shifted reconstruction.

10 Devil's-advocate review

1. **Objection: The centered secant theorem already proves H_N. UPHELD against that conclusion.** The theorem estimates (4.1), but the rootwise Doob-to-terminal embedding with backward heat, low, paid, trace, and forest companions is not proved. Equation (6.4) displays the remaining atoms.

2. **Objection: The quadratic 2^{j-4k} factor was assigned to the whole Gram increment. UPHELD and repaired.** It belongs only to (3.1). The mixed scalar fixture (3.5) proves that the full secant begins at 2^{j-2k}; Theorem 4.1 pays that weaker factor directly.

3. **Objection: The a_k^2 estimate silently uses a higher moment of an adapted selector. DISMISSED.** Equation (4.8) is interpolation on $\Omega \times \mathbb{T}^3$, and (4.9) uses only source L^2 and the accepted terminal sixth moment. No pathwise Bernstein step appears.

4. **Objection: Partial completion spends the endpoint square twice. DISMISSED for Theorem 5.2, upheld for the naive full assembly.** Equation (6.4) labels the fraction θ used by (5.10) and leaves exactly $1-\theta$ in (6.6). Any later proof must retain that reduced coefficient.

5. **Objection: The fresh derivative bound controls $T_G^{>}$. UPHELD against that identification.** Lemma 5.1 is an isolated unshifted form. The shifted adapted coefficient in (6.7) requires a genuine expectation-inside multiplier or cancellation theorem.

6. **Objection: Correcting the zero-slack rows proves Nelson. UPHELD against that conclusion.** Positive arithmetic slack is only a budget check. R-093 requires the full uniform shifted inequality before R-087 CORE can be invoked.

7. **Objection: The revisit example falsifies the full action. DISMISSED.** It falsifies absolute per-increment sixth-moment summation. The terminal shift and the signed outgoing/return contribution cancel, so it supplies no negative divergence of the complete action.

11 Result footer

```
Result ID:              A13-CLASSII-ROOT-LOCAL-GRAM-SECANT-FEEDBACK-BOUNDARY
Ledger:                 R-094.
Claim:                  A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement:      Quadratic curvature has scale 2^(j-4k); the complete
                        centered Gram secant has mixed scale 2^(j-2k) and
                        bound C2^-j0 X^1/2(1+Y)^1/6 with slack 1/3. The fresh
                        derivative slack is 1/6; the value--heat prefix costs
                        O(theta^-1 2^(-3j0)X) from a declared square fraction.
Scope:                  Regular orthogonal strict-past one-shot controls;
                        product L2--L6 interpolation; fixed positive floor.
Dependencies:           R-063, R-066, R-074, R-076, R-079, R-086, R-092, R-093.
Evidence grade:         ANALYTIC, EXACT, EXECUTED, BOUNDARY.
Reproduction command:   python codes/foundations/a13_classii_root_local_
                        gram_secant_feedback_boundary_verify.py
Expected output:        Manifest-pinned primary, independent, integrated, and
                        aggregate assertions PASS; exit zero.
Negative results:       root-factor/square-allocation and BG-row-scope audits;
                        absolute-revisit-sum no-go (all dated 2026-07-27).
Falsification gate:     Failure of any dyadic, Hardy, interpolation, Hermite,
                        prefix, split, revisit, hash, independent, PDF,
                        public-link, count, or scope assertion.
Tier before / after:    T4 / T4; no promotion.
No-overclaim statement: Complete rootwise packet embedding, H_N, REG, and
                        progressive/revisit control, OVERLAP_src, Nelson,
                        cutoff/floor removal, a measure, Sector A, T5--T7 open.
Next required action:   Prove (6.9), including T_G^>, conditional mean debts,
                        and the once-only R-079/R-086 heat--low--trace--forest
                        embedding; then assemble OVERLAP_src.
```